Image-Native Language Models: Reading and Writing as Visual Generation

Imagized Language Model Project

June 2026

Abstract

Most large language models represent language through symbolic text tokens and treat images of writing as an auxiliary input modality. This paper sketches a different research direction: an image-native language model whose inputs and outputs are visual canvases. The model learns from printed pages, handwritten forms, oracle-bone glyphs, bronze inscriptions, seal forms, and other historical scripts as images. It answers by generating readable page images containing modern language, classical-style text, and historically grounded glyph panels. A machine-readable text layer is optional and only applies to spans that computer codecs can represent. We propose a 3090-friendly staged architecture based on visual autoencoding, masked visual language modeling, latent diffusion, and retrieval-augmented glyph evolution.

1 Motivation

Human readers do not see byte-pair tokens. They see pages, lines, strokes, spacing, damage, font variation, and script-specific visual forms. Current language models are extremely capable, but their central representation is normally a sequence of text tokens. Multimodal models accept images, and modern APIs also expose image-generation endpoints. For example, OpenAI documents APIs for processing images as input and generating images as output, and Google's Gemini API documentation describes image generation from text and image prompts [4, 5, 6]. These systems are important baselines, but they do not by themselves define a language model whose primary language substrate is the visible page.

The Imagized Language Model (ILM) goal is different:

$$\text{image input} \rightarrow \text{visual language state} \rightarrow \text{image output.}$$

The output should itself be a readable answer image. For a query such as "explain the evolution of character yan", the system should produce a page containing modern explanatory writing, classical-style Chinese, and oracle, bronze, seal, and modern forms. Typed text can still be accepted by a user interface, but it is rasterized into an image prompt before entering the core model. Conversely, the model's answer is a page image first; OCR or transcription is an optional downstream extraction step.

2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. Token-free language

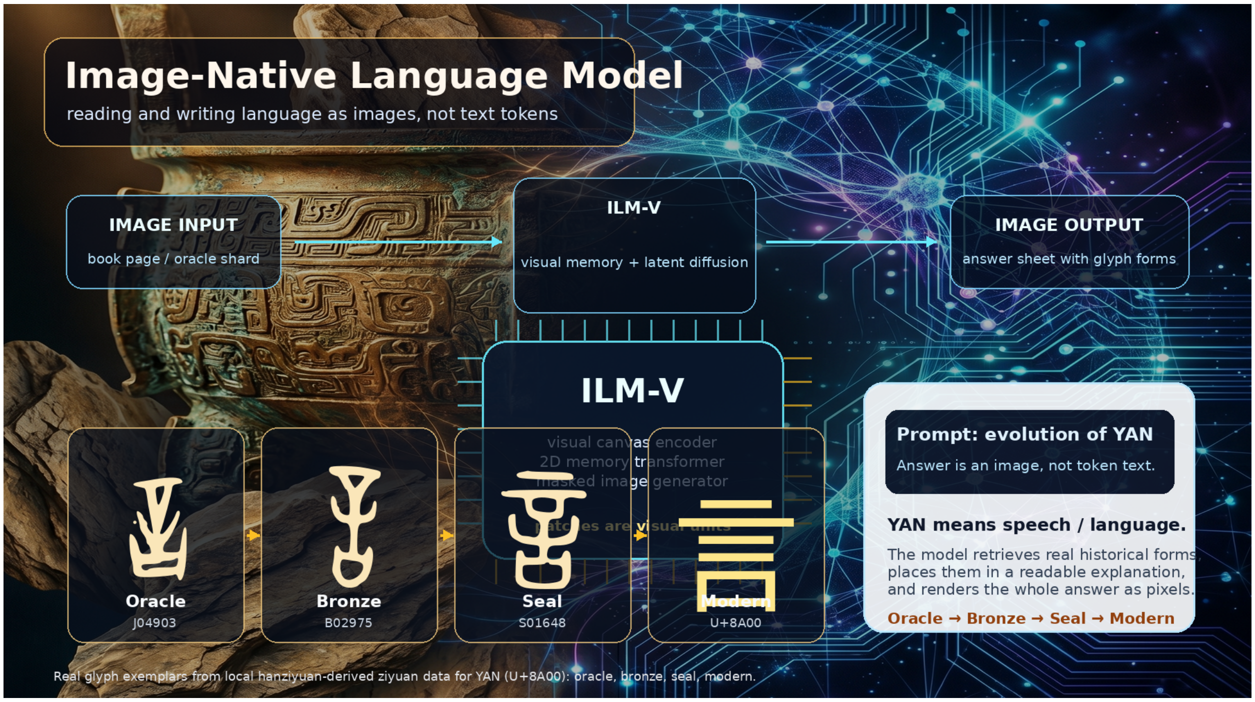

Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for `YAN` (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

models such as CANINE, ByT5, and MEGABYTE show that subword tokenization is not a hard requirement for language modeling. Discrete diffusion language models, including D3PM, Diffusion-LM, SEDD, and LLaDA-style masked diffusion, show that generation need not be left-to-right. Image generative models such as latent diffusion, MaskGIT, and DiT show that high-quality image generation can be performed in compressed latent spaces.

The missing combination is direct: train a model to read and write language as images, while preserving the visual structure of scripts and historical glyphs.

3 Architecture

Figure 2 gives the proposed system. The main components are:

1. **Visual encoder.** A patch/conv hybrid converts glyph tiles, line strips, and page crops into visual latent grids.
2. **Visual memory transformer.** A 2D Transformer maintains cross-panel state with relative position and sparse/window attention.
3. **Latent generator.** A masked diffusion or MaskGIT-like prior fills an output canvas conditioned on input canvas latents.
4. **Image decoder.** A VAE/VQ-VAE decoder converts latents back to readable output images.
5. **Auxiliary readers and critics.** OCR/readability heads evaluate the output, but should not become the primary hidden path.

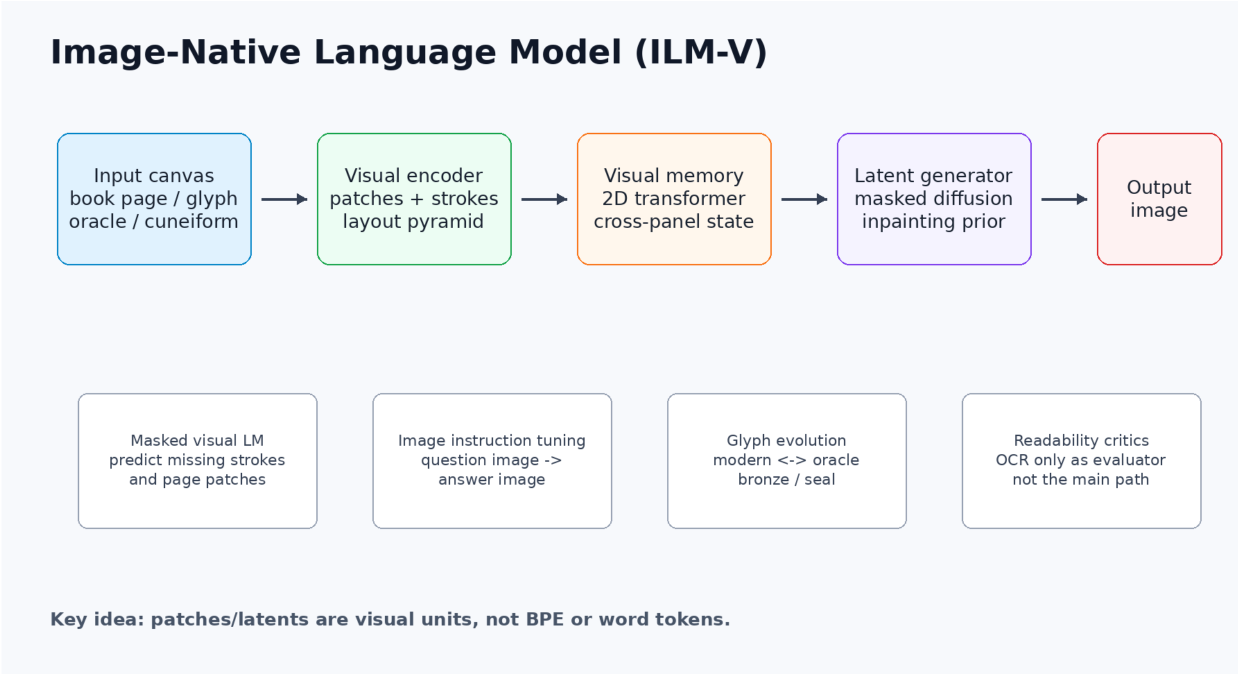

Figure 2: ILM-V architecture. The model consumes input canvases, encodes them into visual latents, maintains a 2D visual memory, and denoises or fills output image latents. OCR-style readers are only auxiliary evaluators, not the main generation path.

The internal units are visual patches or visual codebook entries, not BPE, wordpiece, or Unicode language tokens.

4 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This is enough to train a first glyph autoencoder and a character-evolution generator. Modern book/page data can be added by rendering multilingual corpora into images and by ingesting scanned pages where licenses permit.

5 Training Objectives

The first ILM-V model should combine the following losses:

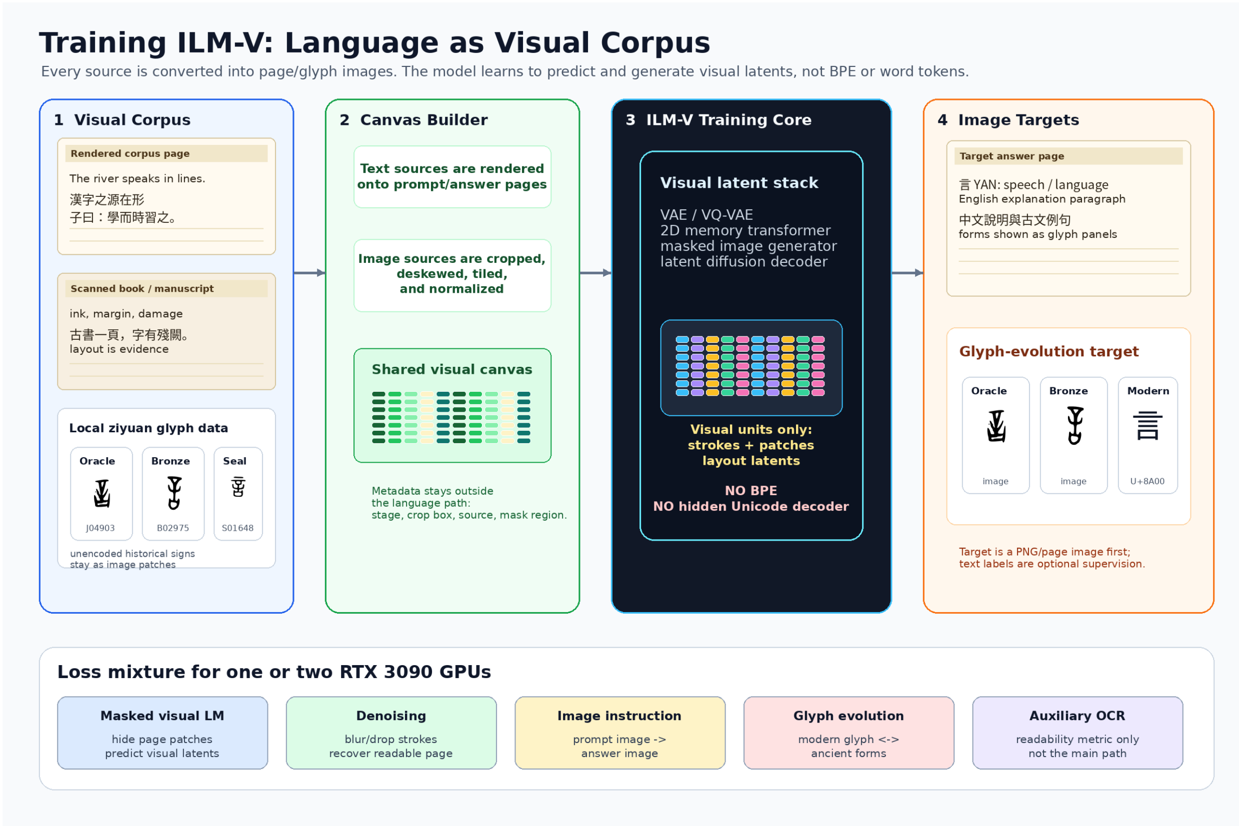

Figure 3: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

1. **Masked visual language modeling:** mask image patches and predict missing visual latents.
2. **Denoising reconstruction:** recover pages and glyphs from blur, noise, damage, dropout, and partial occlusion.
3. **Image-to-image instruction tuning:** render prompts and target answers as images and train conditional generation.
4. **Glyph evolution modeling:** map modern forms to historical stages and historical forms back to modern answer sheets.
5. **Contrastive visual semantics:** align different visual renderings of the same content without making OCR the bottleneck.

6 Image-First Training and Inference

Figure 3 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

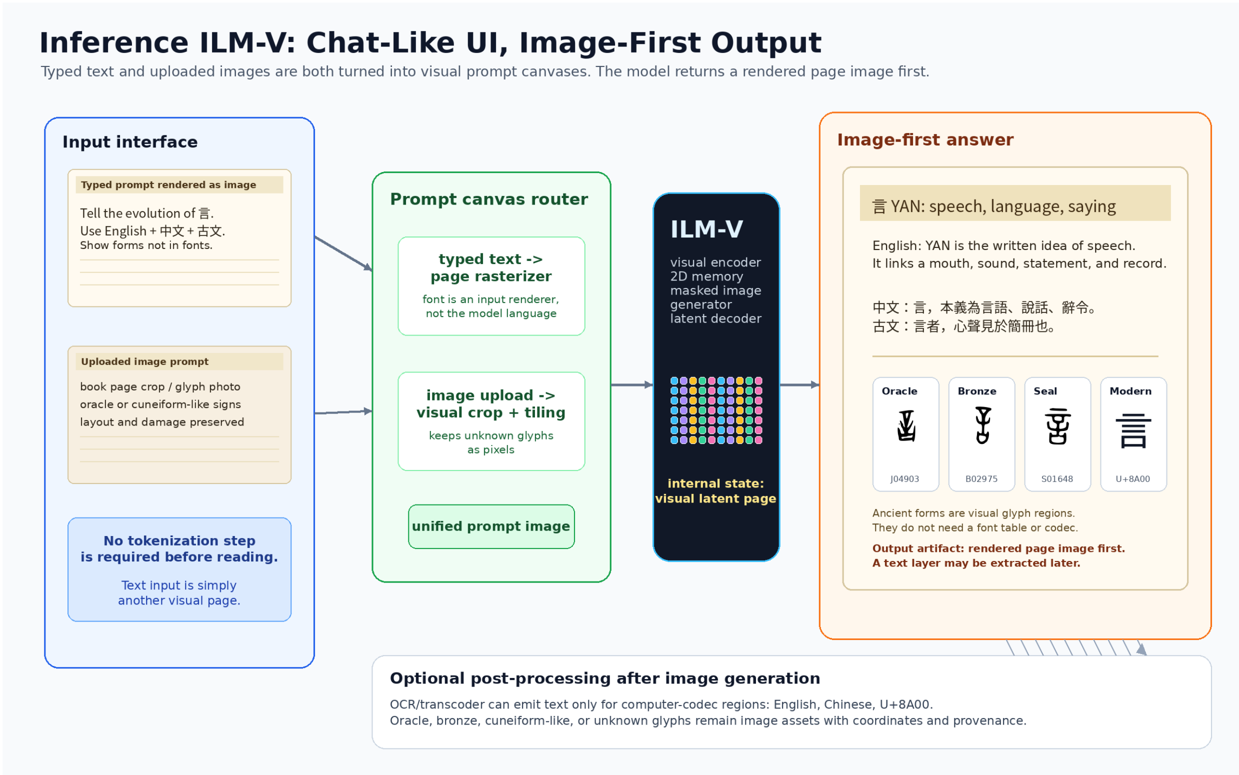

Figure 4: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

Figure 4 shows the user-facing path. A typed prompt is converted into a visual prompt canvas, and an uploaded image is normalized into the same visual space. The model returns a page-like image containing modern language and historical glyph regions. A later OCR/transcoder can attach a searchable text layer for English, Chinese, or Unicode-representable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$
\begin{aligned}
r_\alpha(\text{text prompt}) &\to x_{\text{prompt}}, \\
g_\beta(\text{uploaded page or glyph}) &\to x_{\text{prompt}}, \\
f_\theta(x_{\text{prompt}}, x_{\text{context}}) &\to y_{\text{page}}, \\
o_\psi(y_{\text{page}}) &\to \text{optional text layer} + \text{glyph-region metadata}.
\end{aligned}
$$

Here r_α is only a UI rasterizer, not the language model itself. The core model f_θ maps visual inputs to a rendered page image y_{page}. The optional reader o_ψ may recover Unicode text where possible, but unencoded historical forms remain valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,

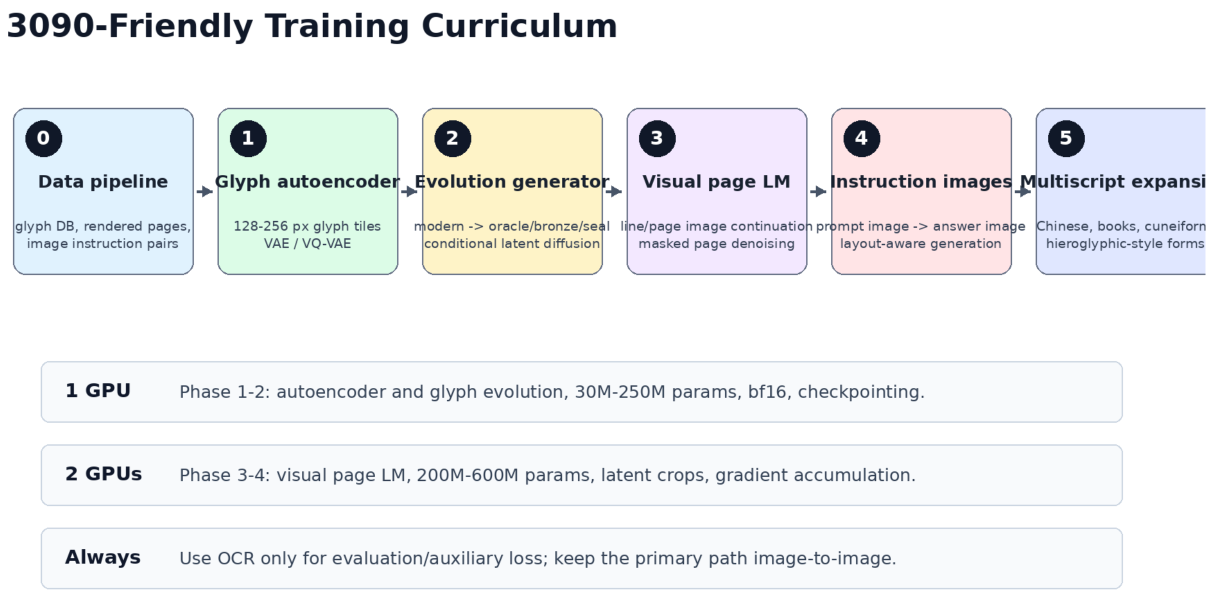

Figure 5: A staged curriculum that fits one or two RTX 3090 GPUs by starting with glyph tiles and latent generation before scaling to page-level image language modeling.

- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from "text answer plus illustrations". The generated page image is the primary output object, and the text layer is secondary.

7 3090-Friendly Curriculum

Figure 5 shows the practical schedule:

1. Build manifests for glyphs, rendered line images, and rendered instruction pairs.
2. Train a 30M–100M parameter glyph/page autoencoder.
3. Train a 100M–250M parameter conditional glyph evolution generator.
4. Train a 200M–600M parameter visual page language model on latent crops.
5. Add instruction tuning with image prompts and image answers.
6. Expand to additional scripts: cuneiform-like signs, hieroglyphic forms, Arabic, Japanese, Korean, and multilingual book pages.

Mixed precision, gradient checkpointing, latent crops, LoRA/adapters, and gradient accumulation make this path realistic on one or two 24GB GPUs.

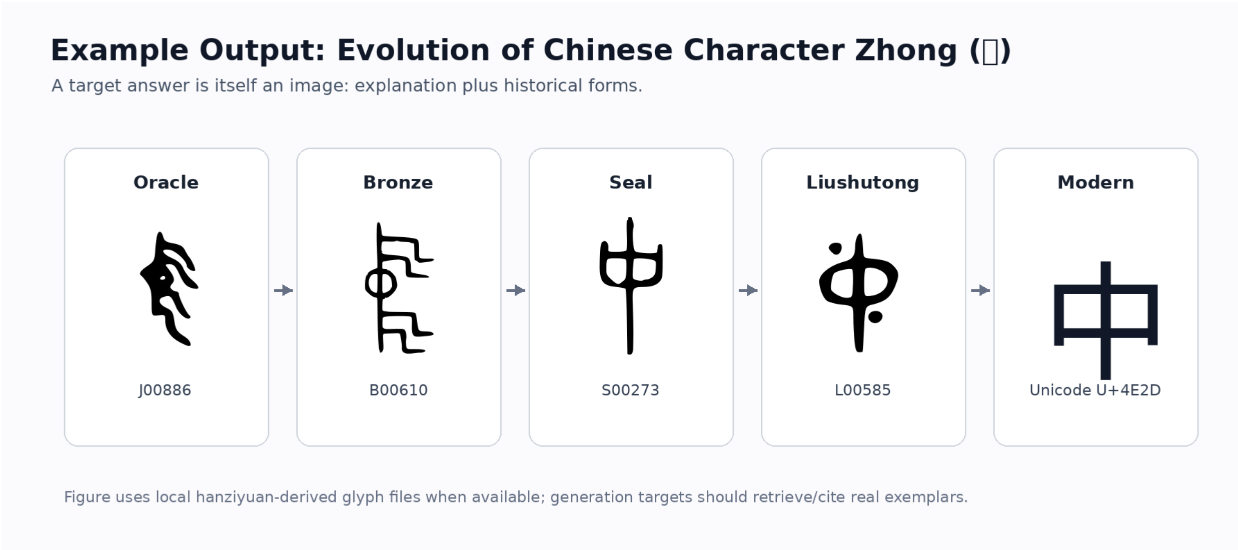

Figure 6: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

8 Target Demonstration

The first demo should answer a rendered visual prompt such as: "Explain the evolution of the Chinese character yan." The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as "yan is the visible trace of speech",
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 6 is a compact stage-timeline example; the Yan figures in Figures 1 and 4 show the fuller answer-page style. Together they test whether the model can combine language understanding, visual page composition, and historical script generation.

9 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

10 Evaluation

The model should be evaluated as both a language model and a visual generator:

- OCR readability of generated modern text images, used only as an external metric.

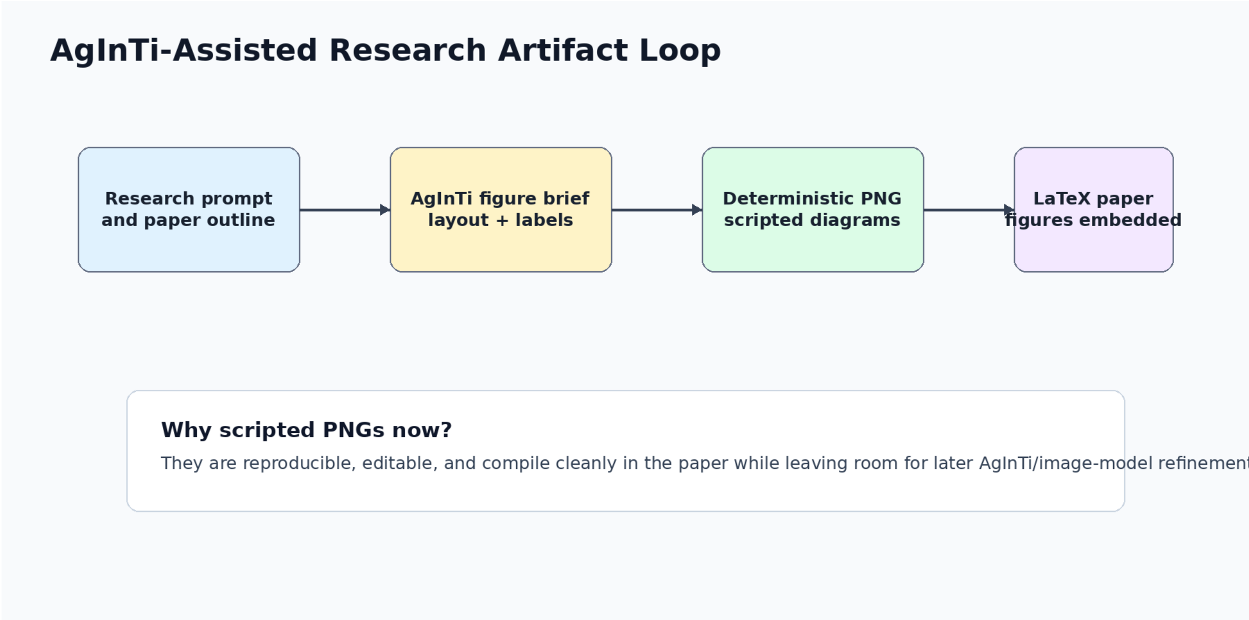

Figure 7: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

- Human readability and task-success ratings.
- Glyph-stage retrieval accuracy: generated glyphs should retrieve correct characters and stages.
- Historical plausibility: distance to nearest real glyph exemplars in stage-specific embedding space.
- Layout validity: no text/panel overlap, stable reading order, correct answer composition.
- Image-to-image QA accuracy on rendered prompts and rendered answers.

11 Conclusion

ILM-V is a feasible research direction because the pieces exist: OCR-free visual readers, token-free language models, visual latent autoencoders, masked image generators, and diffusion language models. The contribution is to make visible writing itself the modeling substrate. The local Chinese etymology dataset gives a concrete first domain where image output is essential rather than decorative.

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